

INSTITUTO TECNOLÓGICO DE SALTILLO

Computo en la Nube

5.10 Actividad: Migración a servicios administrados en Azure

MARIO IRIGOYEN ESTRADA

08 DE MAYO DE 2026

Evidencias para entregar

1. Captura del servidor MySQL Flexible Server.

```
Home
mysql-crud-marioirigoyen
Azure Database for MySQL flexible server

Search  Connect  View process list  Delete  Reset password  Restore  Restart  Upgrade  Stop
2020-03-09 03:12:15.2890233 UTC

Overview
Activity log
Add or remove Favorites by pressing Ctrl+Shift+F

Tags (edit)
Add tags

Getting started  Properties  Recommendations  Monitoring  Tutorials

Switch to PowerShell  Restart  Manage files  New session  Editor  Web preview  Settings  Help

Requesting a Cloud Shell. Succeeded.
Connecting terminal...

Welcome to Azure Cloud Shell

Type "az" to use Azure CLI
Type "help" to learn about Cloud Shell

Subscription used to launch your CloudShell f695d551-013c-4914-bb47-02adcc961c45 is not registered to Microsoft.CloudShell Names
Please follow these instructions "https://aka.ms/RegisterCloudShell" to register. In future, unregistered subscriptions will ha
stricted access to CloudShell service.

Your Cloud Shell session will be ephemeral so no files or system changes will persist beyond your current session.
mario [ ~ ]$ wget --trust-server-names --no-check-certificate "https://go.microsoft.com/fwlink/?linkid=2281474" -O MysqlflexGlob
tCA.crt.pem
MysqlflexGlobalRootC 100% [=====>] 986 --.-
[Files: 1 Bytes: 986 [2.71KB/s] Redirects: 1 Todo: 0 Errors: 0 ]
mario [ ~ ]$ mysql -h mysql-crud-marioirigoyen.mysql.database.azure.com -u admincrud -p "--ssl-ca=MysqlflexGlobalRootCA.crt.pem"
Enter password:
Welcome to the MariaDB monitor. Commands end with ; or \g.
Your MySQL connection id is 13
Server version: 8.0.44-azure Source distribution

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MySQL [(none)]>
```

Flexible server

Microsoft

MySQL

Basics (Change)

Subscription	Azure for Students
Resource group	rg-crud-marioirigoyen
Server name	mysql-crud-marioirigoyen
Administrator login	admincrud
Location	Mexico Central
Availability zone	No preference
High availability	Enabled
High availability mode	Zone redundant
Standby availability zone	No preference
MySQL version	8.0
Compute + storage	General Purpose, D4ds_v4, 4 vCores, 16 GiB RAM, 64 storage, Auto scale IOPS
Backup retention period (in days)	7 day(s)
Accelerated Logs	Not enabled
Storage autogrow	Enabled
Geo-redundancy	Not enabled
Zonal Resiliency	Yes

Networking (Change)

Connectivity method	Public access (allowed IP addresses) and Private endpoint
Allow public access to this resource through the internet using a public IP address	Yes
Allow public access from any Azure service within Azure to this server	Yes

Estimated costs



Compute	USD 284.70/month
Standard_D4ds_v4 (4 vCores, USD 71.17 per vCore) 4 x 71.17	
Storage	USD 8.83/month
64 GiB (USD 0.14 per GiB) 64 x 0.14	
Auto scale IOPS	Auto scale IOPS is billed on usage in per million request increments. Learn more
High availability	USD 293.53/month
Same zone or Zone redundant high availability	
Backup Retention	Backup retention is billed based on additional storage used for retaining backups. Learn more
Bandwidth	Outbound data transfer across services in

Home > Select Azure Database for MySQL deployment option

Flexible server

Microsoft

⚠️ Server names, networking connectivity method, zone redundant HA and backup redundancy cannot be changed after server is created. Review these carefully before provisioning.

Connectivity method ⓘ

☒ Public access (allowed IP addresses) and Private endpoint

☐ Private access (VNet Integration)

ℹ️ Connections from the IP addresses configured in the Firewall rules section below will have access to this server. By default, no public IP addresses are allowed. [Learn more](#)

Database port * ⓘ

3306

Public access

☒ Allow public access to this resource through the internet using a public IP address ⓘ

Firewall rules ⓘ

Inbound connections from the IP addresses specified below will be allowed to port 3306 on this server. [Learn more](#)

☒ Allow public access from any Azure service within Azure to this server ⓘ

+ Add current client IP address (187.161.83.177) + Add 0.0.0.0 - 255.255.255.255

Firewall rule name	Start IP address	End IP address
ClientIPAddress_2026-5-8_21...	187.161.83.177	187.161.83.177
Firewall rule name	Start IP address	End IP address

Private endpoints

Estimated costs 

^ Compute USD 284.70/month

Standard_D4ds_v4 (4 vCores, USD 71.17 per vCore) 4 x 71.17

^ Storage USD 8.83/month

64 GiB (USD 0.14 per GiB) 64 x 0.14

^ Auto scale IOPS

Auto scale IOPS is billed on usage in per million request increments. [Learn more](#)

^ High availability USD 293.53/month

Same zone or Zone redundant high availability

^ Backup Retention

Backup retention is billed based on additional storage used for retaining backups. [Learn more](#)

^ Bandwidth

2. Captura de la base de datos creada.

Home

mysql-crud-marioirigoyen  Azure Database for MySQL flexible server

Search  Connect  View process list  Delete  Reset password  Restore  Restart  Upgrade  Stop

2020-05-09 03:12:13.2890255 UTC

Overview Tags ([edit](#))
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Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

```
MySQL [(none)]> CREATE DATABASE crudusuario CHARACTER SET utf8mb4 COLLATE
-> utf8mb4_unicode_ci;
Query OK, 1 row affected (0.052 sec)
```

```
MySQL [(none)]> show databases;
+-----+
| Database |
+-----+
| crudusuario |
| information_schema |
| mysql |
| performance_schema |
| sys |
+-----+
5 rows in set (0.024 sec)
```

```
MySQL [(none)]> use crudusuario;
Database changed
MySQL [crudusuario]> CREATE TABLE usuario (
-> id INT NOT NULL AUTO_INCREMENT,
-> eMail VARCHAR(150) NOT NULL UNIQUE,
-> pwd_ VARCHAR(255) NOT NULL,
-> fecha DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,
-> estatus CHAR(1) NOT NULL DEFAULT 'A',
-> PRIMARY KEY (id)
-> ) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;
Query OK, 0 rows affected (0.106 sec)
```

3. Captura de la tabla usuarios.

```
Home
mysql-crud-marioirigoyen
Azure Database for MySQL flexible server

Search
Connect View process list Delete Reset password Restore Restart Upgrade
2026-05-09 03:12:13.2890255 UTC

Overview
Activity log
Add or remove favorites by pressing Ctrl+Shift+F
Tags (edit)
Add tags
Getting started Properties Recommendations Monitoring Tutorials

Switch to PowerShell Restart Manage files New session Editor Web preview Settings Help

-> pwd_ VARCHAR(255) NOT NULL,
-> fecha DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,
-> estatus CHAR(1) NOT NULL DEFAULT 'A',
-> PRIMARY KEY (id)
-> ) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;
Query OK, 0 rows affected (0.106 sec)

MySQL [crudusuario]> insert into usuarios values ('pruebaTerminal', 'prueba@email.com', 'contra')
-> ;
ERROR 1146 (42S02): Table 'crudusuario.usuarios' doesn't exist
MySQL [crudusuario]> insert into usuario values ('pruebaTerminal', 'prueba@email.com', 'contra');
ERROR 1136 (21S01): Column count doesn't match value count at row 1
MySQL [crudusuario]> insert into usuario values (1, 'pruebaTerminal', 'prueba@email.com', 'contra');
ERROR 1136 (21S01): Column count doesn't match value count at row 1
MySQL [crudusuario]> insert into usuario values (1, 'prueba@email.com', 'contra');
ERROR 1136 (21S01): Column count doesn't match value count at row 1
MySQL [crudusuario]> INSERT INTO usuario (eMail, pwd_, fecha, estatus)
-> VALUES ('pruebaTerminal@email.com', '1234', NOW(), 'A');
Query OK, 1 row affected (0.061 sec)

MySQL [crudusuario]> select * from usuarios;
ERROR 1146 (42S02): Table 'crudusuario.usuarios' doesn't exist
MySQL [crudusuario]> select * from usuario;

+-----+-----+-----+-----+
| id | eMail | pwd_ | fecha | estatus |
+-----+-----+-----+-----+
| 1 | pruebaTerminal@email.com | 1234 | 2026-05-09 03:31:53 | A |
+-----+-----+-----+-----+
1 row in set (0.022 sec)

MySQL [crudusuario]>
```

Transfiriendo info de la bd pasada

```
Database changed
MySQL [crudusuario]> exit
Bye
mario [ ~ ]$ mysql -h mysql-crud-nombreadapellido.mysql.database.azure.com -u admincrud -p
crudusuario < crudusuario.sql
Enter password:
ERROR 2005 (HY000): Unknown server host 'mysql-crud-nombreadapellido.mysql.database.azure.com' (-2)
bash: crudusuario: command not found
mario [ ~ ]$ mysql -h mysql-crud-marioirigoyen.mysql.database.azure.com -u admincrud -p crudusuario < crudusuario.sql
Enter password:
ERROR 2005 (HY000): Unknown server host 'mysql-crud-nombreadapellido.mysql.database.azure.com' (-2)
mario [ ~ ]$ mysql -h mysql-crud-marioirigoyen.mysql.database.azure.com -u admincrud -p crudusuario < crudusuario.sql
Enter password:
mario [ ~ ]$ mysql -h mysql-crud-nombreadapellido.mysql.database.azure.com -u admincrud -p
Enter password:
ERROR 2005 (HY000): Unknown server host 'mysql-crud-nombreadapellido.mysql.database.azure.com' (-2)
mario [ ~ ]$ mysql -h mysql-crud-marioirigoyen.mysql.database.azure.com -u admincrud -p "--ssl-ca=MySqlflexGlobalRootCA.crt.pem"
Enter password:
Welcome to the MariaDB monitor. Commands end with ; or \g.
Your MySQL connection id is 16
```


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Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MySQL [(none)]> show tables;

ERROR 1046 (3D000): No database selected

MySQL [(none)]> use crudusuario

Reading table information for completion of table and column names

You can turn off this feature to get a quicker startup with -A

Database changed

MySQL [crudusuario]> show tables;

-----+
| Tables_in_crudusuario |
-----+
| usuario |
-----+

-----+

-----+

-----+

-----+

1 row in set (0.022 sec)

MySQL [crudusuario]> select * from usuario;

-----+-----+-----+-----+
| id | eMail | pwd_ | fecha | estatus |
-----+-----+-----+-----+
| 1 | marioirigoyen24@outlook.com | contrasena | 2026-05-01 00:00:00 | A |
| 2 | pruebatarea | hola, no encripta? | 2026-05-01 00:00:00 | B |
-----+-----+-----+-----+

-----+

-----+

-----+

-----+

-----+

2 rows in set (0.021 sec)

MySQL [crudusuario]>

MySQL [crudusuario]> select * from usuarios;

ERROR 1146 (42S02): Table 'crudusuario.usuarios' doesn't exist

MySQL [crudusuario]> select * from usuario;

-----+-----+-----+-----+
| id | eMail | pwd_ | fecha | estatus |
-----+-----+-----+-----+
| 1 | pruebaTerminal@email.com | 1234 | 2026-05-09 03:31:53 | A |
-----+-----+-----+-----+

-----+

-----+

-----+

-----+

-----+

1 row in set (0.022 sec)

MySQL [crudusuario]> exit

Bye

mario [~]\$ ls

MysqlflexGlobalRootCA.crt.pem

mario [~]\$ ls

crudusuario.sql MysqlflexGlobalRootCA.crt.pem

mario [~]\$

~ took 4m4s

> scp AppServer:crudusuario.sql
crudusuario.sql

100% 2307 16.9KB/s

~

> ls

Android Documents go Projects tmp winboat kubectl.sha256
config-bak Downloads Music Public Videos crudusuario.sql restoreHyprlock.sh
Desktop Games Pictures Templates 'VirtualBox VMs' kubectl

~

>


```

No entry for terminal type "xterm-kitty";
using dumb terminal settings.
No entry for terminal type "xterm-kitty";
using dumb terminal settings.
Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> show database
→ ;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MariaDB server version
ntax to use near 'database' at line 1
MariaDB [(none)]> show databases;
+-----+
| Database |
+-----+
| cnube    |
| information_schema |
| mysql    |
| performance_schema |
| sys      |
+-----+
5 rows in set (0.005 sec)

MariaDB [(none)]> exit
Bye
azureuser@AppServer:~$ mysqldump -u root -p cnube > crudusuario.sql
Enter password:
azureuser@AppServer:~$ ls
crudusuario.sql
azureuser@AppServer:~$

```

4. Captura del App Service.

Publish * ☒ Code ☐ Container

Runtime stack *

Operating System * ☒ Linux ☐ Windows

Region *

 Not finding your App Service Plan? Try a different region or select your App Service Environment.

Pricing plans

App Service plan pricing tier determines the location, features, cost and compute resources associated with your app. [Learn more](#)

Linux Plan (Mexico Central) *



[Create new](#)

Pricing plan

[Explore pricing plans](#)

Zone redundancy

An App Service plan can be deployed as a zone redundant service in the regions that support it. Your initial instance count will be set based on your zone redundancy configuration. To ensure you'll be able to enable zone redundancy at any point in the lifecycle of your app, enable zone redundancy now. You can decrease the instance count after the App Service plan is created. [Learn more](#)

Zone redundancy

- ☐ **Enabled:** Your App Service plan and the apps in it will be zone redundant. The minimum App Service plan instance count will be two.
- ☒ **Disabled:** Your App Service plan and the apps in it will not be zone redundant. The minimum App Service plan instance count will be one.

[Review + create](#)

[< Previous](#)

[Next : Database >](#)

App Service Web Apps lets you quickly build, deploy, and scale enterprise-grade web, mobile, and API apps running on any platform. Meet rigorous performance, scalability, security and compliance requirements while using a fully managed platform to perform infrastructure maintenance. [Learn more](#)

Project Details

Select a subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription * ⓘ Azure for Students ▼

Resource Group * ⓘ rg-crud-marioirigoyen ▼
[Create new](#)

Instance Details

Name app-crud-marioirigoyen ✓
-eddzaceag5dxjdf.mexicocentral-01.azurewebsites.net

☒ Secure unique default hostname on. [More about this update](#)

Publish * ☒ Code ☐ Container

Runtime stack * PHP 8.5 ▼

Operating System * ☒ Linux ☐ Windows

Region * Mexico Central ▼

i Not finding your App Service Plan? Try a different region or select your App Service Environment.

Pricing plans

App Service plan pricing tier determines the location, features, cost and compute resources associated with your app. [Learn more](#)

[Review + create](#) [< Previous](#) [Next : Database >](#)

5. Captura de variables de entorno configuradas, ocultando contraseña.

The screenshot shows the Microsoft Azure portal interface. The top navigation bar includes the Azure logo, a search bar, and the user profile 'L21051450@saltill...'. The main content area is titled 'app-crud-marioirigoyen | Environment variables'. It features a table of environment variables:

Name	Value	Deployment slot sett...	Source	Delete
DB_HOST	Show value		App Service	
DB_NAME	Show value		App Service	
DB_PASS	Show value		App Service	
DB_USER	Show value		App Service	

Below the table, there is a code editor for 'Database.php'. The code is as follows:

```

1 <?php
2
3 reference | 0 implementations
4 class Database
5 {
6     1 reference | 0 overrides
7     public static function StartUp()
8     {
9         $host = getenv("DB_HOST");
10        $db = getenv("DB_NAME");
11        $user = getenv("DB_USER");
12        $pass = getenv("DB_PASS");
13
14        // $pdo = new PDO("mysql:host=172.16.10.156;port=3306;dbname=cNube;charset=utf8", 'msb', 'admin. ');
15        // $pdo->setAttribute(PDO::ATTR_ERRMODE, PDO::ERRMODE_EXCEPTION);
16
17        $pdo = new PDO("mysql:host=$host;dbname=$db;charset=utf8mb4", $user, $pass, [
18            PDO::ATTR_ERRMODE => PDO::ERRMODE_EXCEPTION,
19            PDO::ATTR_DEFAULT_FETCH_MODE => PDO::FETCH_ASSOC,
20        ]);
21
22        return $pdo;
23    }
24 }

```

On the right side of the image, there is a document titled 'file:///home/tarmairon/Dov' with the following text:

Settings → Environment variables
 O en algunos portales:
 Configuration → Application settings
 En App Service, las configuraciones de aq
 de entorno.

Paso 18. Agregar variables

Agregar:

NOMBRE	Valor
DB_HOST	mysql-crud-nombre
DB_NAME	crudusuario
DB_USER	admincrud
DB_PASS	MySql.Azure.2026!

Guardar cambios.
 Presionar:
 Apply
 Aceptar reinicio si Azure lo solicita.

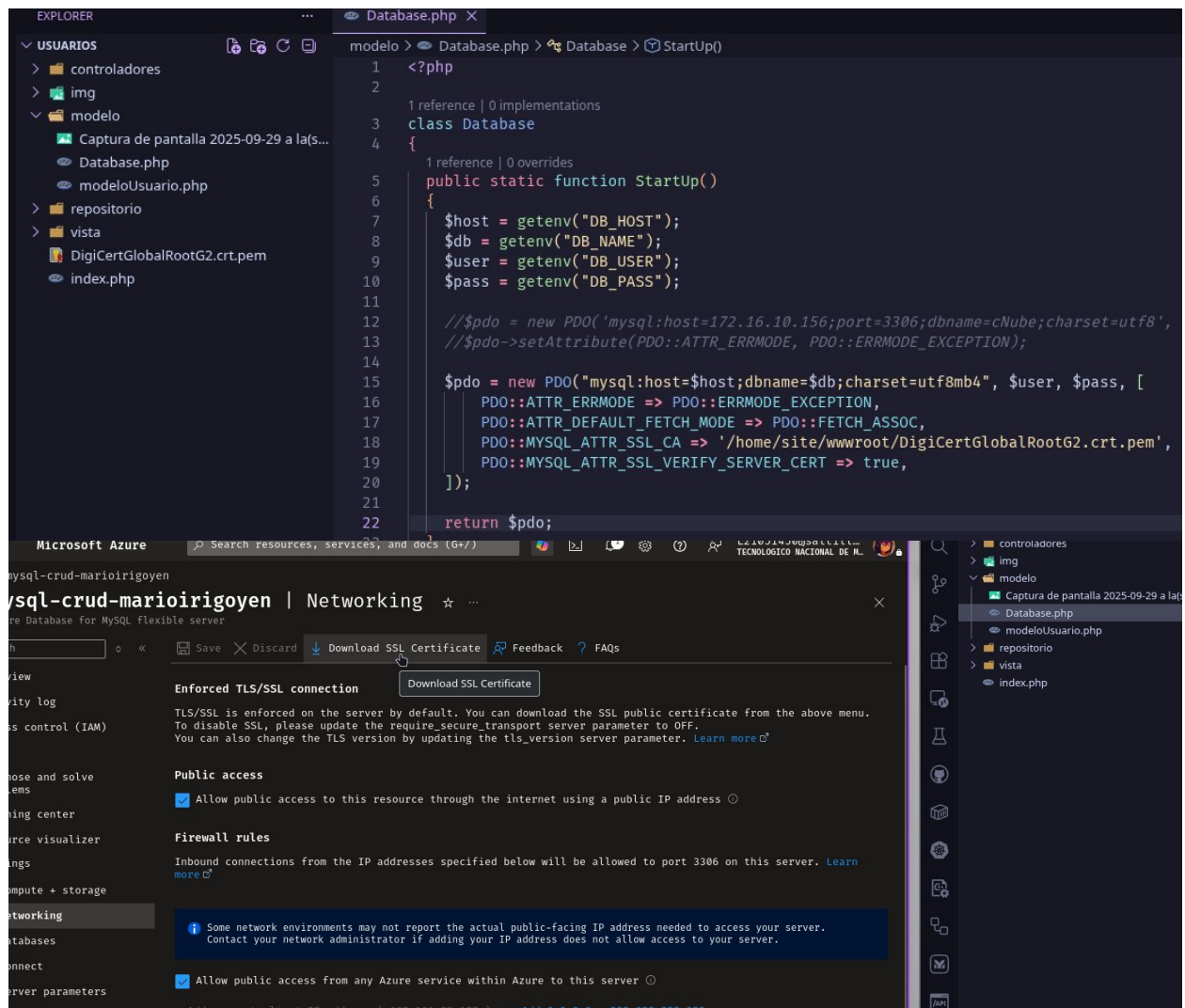
The screenshot shows the 'Publish files' dialog in the Azure portal. The 'Source' dropdown is set to 'Publish files (new)'. The text below reads:

Publish files

Choose your ZIP package (maximum file size 2 gigabytes) using the form below. Then, click "Save" to upload. The site will automatically restart once the upload is complete. You can monitor progress in the "Logs" section. [Learn more](#)

The input field contains 'usuarios.zip' and there is a 'Browse' button next to it.

No podia por que faltaba el certificado del mysql flexible server, se descargo y se pudo en el zip y se agrego a que lo use el codigo



6. Captura de la aplicación funcionando.

← → ↻ app-crud-marioirigoyen-eddzaceag5dxjdf.mexicocentral-01.azurewebsites.net/cont ⚙ 🔑 ⌚ ⬇ 🔦 📄

Mostrando Registros de Usuarios

[Agregar](#) / [Salir](#)

Show entries Buscar

eMail	Fecha	Estatus		
marioirigoyen24@outlook.com	2026-05-01 00:00:00	A		
pruebatarea	2026-05-01 00:00:00	B		

Mostrando página 1 de 1

7. Captura de una operación INSERT.

← → ↻ app-crud-marioirigoyen-eddzaceag5dxjdf.mexicocentral-01.azurewebsites.net/cont ⚙ 🔑 ⌚ ⬇ 🔦 📄

Agregando Usuario

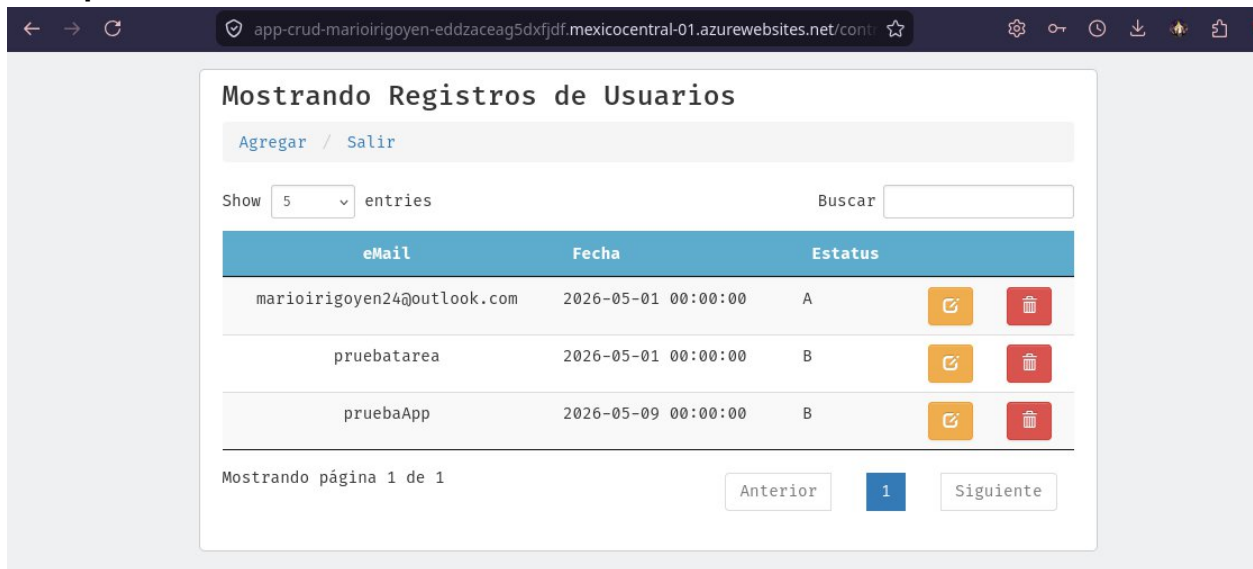
[Salir](#)

Correo Electrónico

Contraseña

Estatus

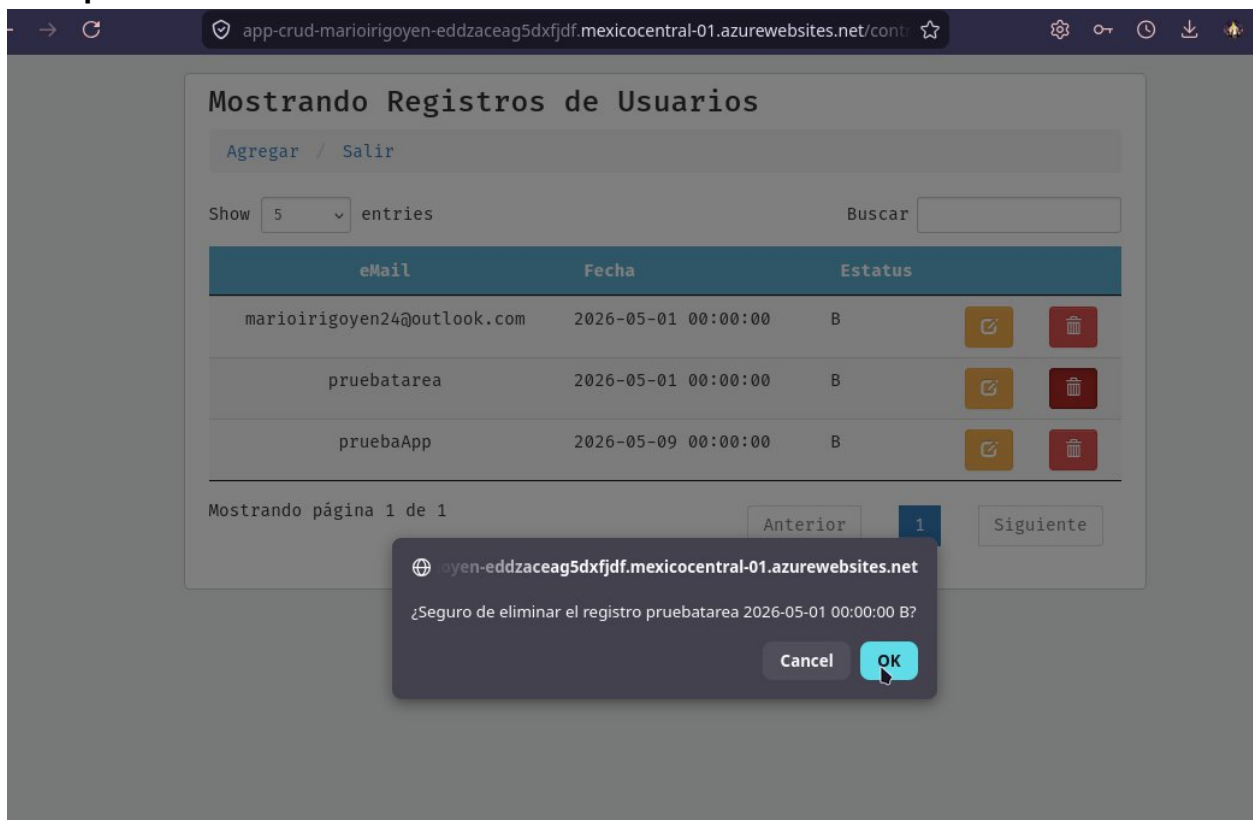
8. Captura de una consulta SELECT.



The screenshot shows a web browser displaying a page titled "Mostrando Registros de Usuarios". The page has a navigation bar with "Agregar" and "Salir" links. Below the navigation bar, there is a "Show" dropdown set to "5" and a "Buscar" search input. The main content is a table with three columns: "eMail", "Fecha", and "Estatus". The table contains three rows of user data. Each row has two action buttons: a yellow "Editar" (Edit) button and a red "Eliminar" (Delete) button. At the bottom, there is a pagination bar showing "Mostrando página 1 de 1" and buttons for "Anterior", "1", and "Siguiete".

eMail	Fecha	Estatus	Editar	Eliminar
marioirigoyen24@outlook.com	2026-05-01 00:00:00	A		
pruebatarea	2026-05-01 00:00:00	B		
pruebaApp	2026-05-09 00:00:00	B		

9. Captura de una modificación o eliminación.



The screenshot shows the same web application as in the previous image, but with a confirmation dialog box overlaid. The dialog box has a title bar with the URL "marioirigoyen-eddzaceag5dxjfdf.mexicocentral-01.azurewebsites.net". The main text of the dialog asks: "¿Seguro de eliminar el registro pruebatarea 2026-05-01 00:00:00 B?". At the bottom of the dialog, there are two buttons: "Cancel" and "OK". The "OK" button is highlighted with a mouse cursor.

eMail	Fecha	Estatus	Editar	Eliminar
marioirigoyen24@outlook.com	2026-05-01 00:00:00	B		
pruebatarea	2026-05-01 00:00:00	B		
pruebaApp	2026-05-09 00:00:00	B		

Mostrando Registros de Usuarios

[Agregar](#) / [Salir](#)

Show entries

Buscar

eMail	Fecha	Estatus		
marioirigoyen24@outlook.com	2026-05-01 00:00:00	B		
pruebaApp	2026-05-09 00:00:00	B		

Mostrando página 1 de 1

Anterior

1

Siguiente

Modificando Registros de Usuario

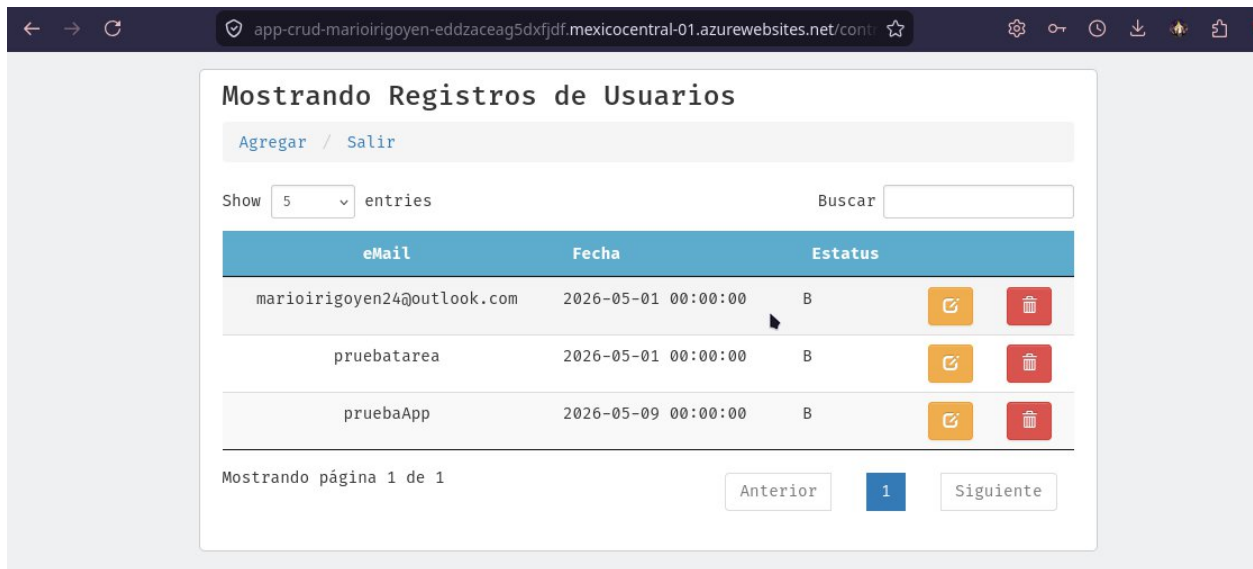
[Salir](#)

Correo Electrónico

Contraseña

Estatus

Guardar



10. Breve explicación de la diferencia entre:

o VM con LAMP

o App Service

o MySQL Flexible Serve

VM con LAMP es infraestructura pura — tú instalas y administras todo: Apache, MySQL, PHP, actualizaciones, seguridad, firewall. Control total pero toda la responsabilidad es tuya. Lo viviste hoy con el ufw, los permisos de www-data, etc.

App Service es PaaS — Azure administra el servidor, el SO, el runtime de PHP. Tú solo subes tu código en un zip y funciona. No tienes SSH al sistema, no configuras Apache, no te preocupas por parches. Escalabilidad automática incluida.

MySQL Flexible Server es DBaaS — base de datos administrada. Azure maneja backups, alta disponibilidad, parches, SSL. Tú solo te conectas con un host/user/pass. Es el equivalente a App Service pero para la base de datos.

Contestar las siguientes preguntas:

1. ¿Qué ventaja tiene usar App Service en lugar de administrar Apache manualmente?

Pues fue tan facil como crearla, subir el zip, y si no hubiera sido por lo del cert, funciona a la primera, ahi si se me paso, o incluso si me hubiera puesto a hacer que el codigo no lo usara, pero bueno, es mucho mas sencillo, administrar apache local te tienes que preocupar del server, de su refrigeracion, en una vm te olvidas de eso pero no de

instalar, actualizar, conectarte y subir los cambios, aqui pues solo pones el zip y ya quedo, o desde github, o con un pipeline aun mejor, ahorra mucho tiempo y te puedes cocentrar en hacer codigo. Aunque me dedico a lo que esto ahorra jaja

2. ¿Qué significa que MySQL sea un servicio administrado?

Que tu nadamas indicas tu usuario y contrase;a y ya microsoft se encarga de lo demas, creas tus schemas y ya no te preocupas por certificados y permisos o usar un usuario que no sea root por que aqui no te deja y asi

3. ¿Por qué no conviene guardar contraseñas directamente en el código PHP?

Por que se pueden ver los archivos desde el navegador o alguien puede entrar al servidor si configuramos algo mal y pues ya tiene todas nuestras credenciales

4. ¿Qué problemas se reducen al usar variables de entorno?

Es mas seguro, si el codigo es publico no hay que preocuparse, no se pueden cambiar utilizando inyecciones de codigo etc

5. ¿Qué diferencias encontró entre instalar todo en una VM y usar servicios

administrados? La facilidad, noma subes tu codigo o base de dato y ya quedo. Te ahorra mucho tiempo y si es como el serverless de aws pues tambien te cobran por el uso y no por que esta prendida todo el tiempo

Conclusion

Pues los servicios administrados o serverless, son en los que el proveedor se encarga de todo, y te facilita mucho la vida, en proyectos peque;os, mvp, o proyectos personales y te puedes centrar a programar en vez de estar configurando servers o bases de datos. Ya en proyectos mas grandes y especializados ya lo mejor y mas barato es usar servers que tu administras o incluso fisicos, todo depende mucho de los recursos, el tipo de cliente, app o solucion que se este dando. Pero si facilitan mucho las cosas, y tambien en lo que es escalamiento y CI\CD